Range of $\alpha-$particles and setting up $\alpha-\gamma$ coincidence SUPPLEMENTARY MATERIAL

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I. CODE AND DATA AVAILABILITY

All the Analysis code, plotting scripts, experimental data, and supplementary material are available in the GitHub repository.

II. SUPPLEMENTARY MATERIAL

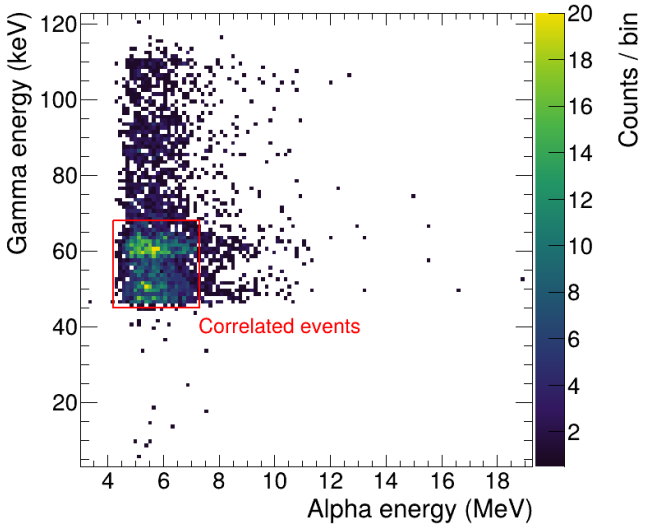


FIG. 1. $\alpha-\gamma$ energy-correlation heatmap on a linear colour scale. The 2D distribution shows the coincidence-event counts as a function of calibrated alpha-particle and gamma-ray energies.

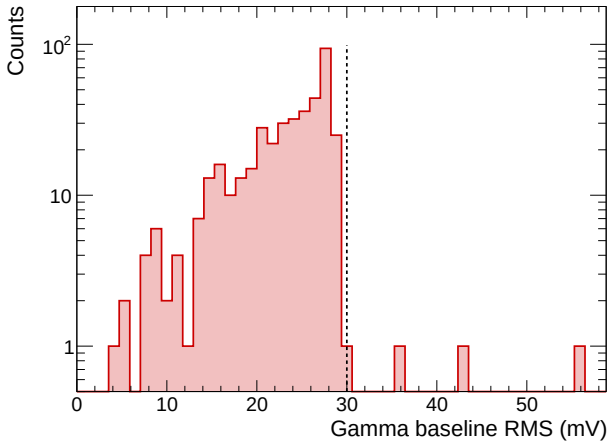


FIG. 2. Distribution of the gamma-signal baseline RMS for the recorded events. The distribution was used to select a baseline-RMS threshold for rejecting noisy events.

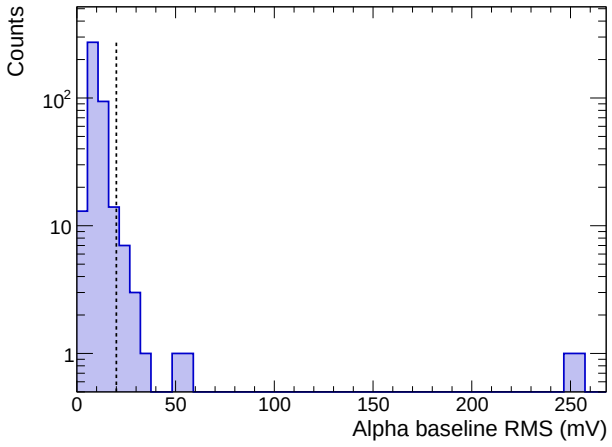


FIG. 3. Distribution of the alpha-signal baseline RMS for the recorded events. The distribution was used to select a baseline-RMS threshold for rejecting noisy events.

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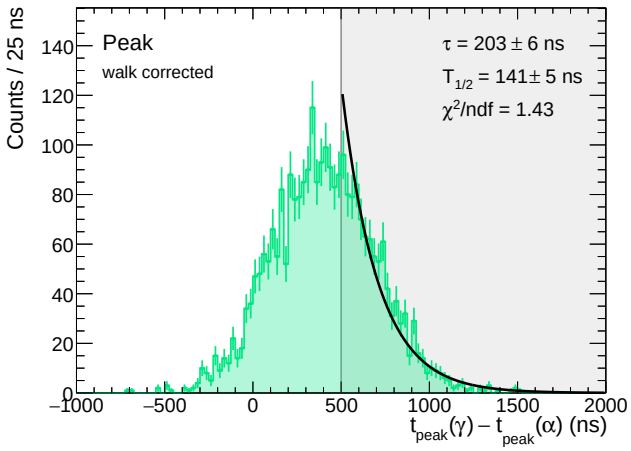


FIG. 4. Time-difference distribution based on pulse-peak times after time-walk correction. The fitted exponential region gives a mean lifetime of (203 ± 6) ns and a corresponding half-life of (141 ± 5) ns.

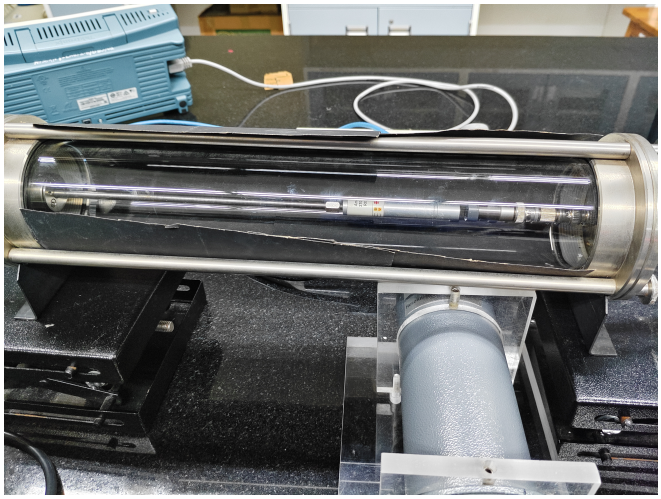


FIG. 5. Close-up of the experimental setup around the alpha detector, showing its placement relative to the radioactive source.



FIG. 6. Oscilloscope used to acquire the alpha and gamma detector signals for the coincidence measurements.